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Internship Reference

Mr Rahul Shailesh Kumar Yadav born on 6 January 2000 was employed in our company from 1 February 2026 to 31 August 2026 in the AD System Development department as an Intern - XiL Testing.

MOIA GmbH is a technology company of the Volkswagen Group focused on building integrated mobility solutions, with a strong emphasis on autonomous driving. The company develops and operates on-demand ride pooling services while also providing a comprehensive, end-to-end platform that includes fleet and driver management, passenger services, and B2B solutions. In addition, MOIA is evolving into a provider of autonomous mobility systems, combining self-driving vehicles, intelligent software, and operational services to enable cities and transport operators to implement scalable, efficient mobility solutions. With approximately 800 employees, MOIA brings together developers, engineers, designers, and strategists working across its offices in Berlin, Hamburg, Munich, Hannover, and Wolfsburg, as well as internationally in the United States. Together, they are dedicated to advancing sustainable, connected, and increasingly autonomous urban transportation.

Within the scope of his duties, Mr Yadav was responsible for these tasks:

- Developed a data-driven vehicle dynamics model to replace the physical vehicle in the Software-in-the-Loop (SiL) test environment.
- Trained a machine learning network (a Long Short-Term Memory recurrent neural network) on real-world fleet data recorded from the vehicle CAN (Controller Area Network) bus to predict vehicle states.
- Integrated the model into the simulation framework as a FEP (Functional Engineering Platform) participant, so it could be used for automated evaluations.
- Evaluated the model in both open-loop and closed-loop operation against the existing physics-based digital twin and against real vehicle recordings.
- Supported the team in analyzing test scenario results from new software releases by developing a LLM-based automated analysis tool.
- Worked as part of an agile development team, contributing to sprint planning, reviews and code reviews.
- Technologies used: Python, Machine Learning Libraries (PyTorch, NumPy), C++, and FEP SDK 3 simulation middleware for distributed co-simulation.

Mr Yadav always acquired very good specialist knowledge within a very short time in the respective departments. His very good technical understanding forms the basis of his ability to present the most complex issues simply and comprehensibly.

Rahul consistently performed his duties with great dedication, initiative, and professionalism. He demonstrated excellent judgement in prioritising tasks and maintained a clear focus on achieving project objectives. His persistence and dependable follow-up ensured that

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important issues were addressed effectively and in a constructive manner.He quickly familiarised himself with complex technical topics and worked independently and efficiently, requiring only minimal guidance. Rahul consistently showed a proactive and solution-oriented mindset, developing his own ideas and identifying effective approaches to overcome challenges.A particularly outstanding accomplishment was the successful integration of a vehicle model into a highly sophisticated simulator environment. Despite the complexity of the task, he completed this work largely independently, demonstrating excellent analytical skills, technical competence, and determination. His contributions had a significant positive impact on the success of the project.Rahul's performance was consistently of a very high standard and he was a valued member of the team. He had an excellent ability to familiarise himself rapidly, which always enabled him to grasp even the most complex work content very well within a very short time. Mr Yadav always worked using his own initiative and completely identified with his responsibilities and our company at all times. He always displayed an impressive level of dedication and motivation. During the entire internship, Mr Yadav stood out because of his extremely high and excellent eagerness to learn. In extremely stressful situations, he always displayed exemplary resilience.

He was always highly appreciated by our company because of his extremely diligent and responsible way of working. Mr Yadav always was extremely reliable.

He always found excellent solutions for any problems which arose. One of Rahul's most significant achievements was his successful contribution to the development and integration of an AI-based vehicle model as part of his master's thesis. The objective of the project was to develop a solution capable of replacing the existing model-based digital twin within our simulation environment.Throughout the project, Rahul demonstrated exceptional self-motivation, technical competence, and determination. He consistently delivered high-quality results while working with a high degree of independence and requiring only minimal supervision. His successful implementation and integration of the AI-based vehicle model constituted a valuable contribution to the project and laid an important foundation for future developments in our simulation environment. The performance of Mr Yadav during his internship was always fully appreciated by us in every respect.

He was highly appreciated both inside and outside the company for his consistently friendly and even-tempered disposition. He was always helpful, courteous and even put his personal interests on hold if necessary. His conduct towards his line managers, supervisors, colleagues or towards clients was always exemplary and loyal. He was friendly and helpful towards other trainees at all times.

We would like to thank Mr Yadav for the consistently very good and pleasant collaboration. We hope the period of training has reinforced his choice of profession and motivated him for the future as we believe he is very well-suited for his chosen profession. For this reason, we have offered him an apprenticeship. We hope that we can continue our collaboration as part of the training scheme. We wish Mr Yadav continued success and all the best for his future career and private life.

Hamburg, 31 August 2026

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